

DETAILED COMPARISON OF PRETRAINED CNN MODELS

This document provides an in-depth structured comparison of the five most influential pretrained deep learning models used in Computer Vision: AlexNet, GoogleNet, ResNet, VGG16 and VGG19.

1. BASIC INFORMATION

- 1 AlexNet – Introduced in 2012 by University of Toronto
- 2 GoogleNet – Introduced in 2014 by Google
- 3 ResNet – Introduced in 2015 by Microsoft Research
- 4 VGG16 – Introduced in 2014 by Oxford VGG Group
- 5 VGG19 – Introduced in 2014 by Oxford VGG Group

2. ARCHITECTURE DEPTH

- 1 AlexNet: 8 layers
- 2 GoogleNet: 22 layers
- 3 ResNet: 50, 101, 152 layer variants
- 4 VGG16: 16 layers
- 5 VGG19: 19 layers

3. NUMBER OF PARAMETERS

- 1 AlexNet: around 60 million
- 2 GoogleNet: around 7 million
- 3 ResNet50: around 25 million
- 4 VGG16: around 138 million
- 5 VGG19: around 144 million

4. MODEL SIZE

- 1 AlexNet: approx 240 MB
- 2 GoogleNet: approx 50 MB
- 3 ResNet50: approx 98 MB
- 4 VGG16: approx 528 MB
- 5 VGG19: approx 549 MB

5. ACCURACY ON IMAGENET

- 1 AlexNet: Top-5 error around 15.3%
- 2 VGG16: Top-5 error around 7.5%
- 3 GoogleNet: Top-5 error around 6.7%
- 4 VGG19: Top-5 error around 7.3%

- 5 ResNet: Top-5 error around 3.6%

6. ARCHITECTURE STYLE

- 1 AlexNet: Traditional sequential CNN
- 2 GoogleNet: Inception module based parallel CNN
- 3 ResNet: Residual learning with skip connections
- 4 VGG16/VGG19: Uniform 3x3 convolution architecture

7. COMPUTATIONAL REQUIREMENTS

- 1 VGG16/VGG19: Very high computation and memory usage
- 2 GoogleNet: Very lightweight and efficient
- 3 ResNet: Balanced computation with high performance
- 4 AlexNet: Moderate resource requirement

8. TRAINING CHARACTERISTICS

- 1 AlexNet: Easy to train but prone to overfitting
- 2 GoogleNet: Efficient training with auxiliary classifiers
- 3 ResNet: Easiest to train deep networks due to skip connections
- 4 VGG models: Harder to train due to large parameters

9. ADVANTAGES

- 1 AlexNet: Simplicity and speed
- 2 GoogleNet: Very lightweight and fast
- 3 ResNet: Extremely high accuracy and scalability
- 4 VGG: Excellent for feature extraction and transfer learning

10. DISADVANTAGES

- 1 AlexNet: Low accuracy by modern standards
- 2 GoogleNet: Complex internal design
- 3 ResNet: Architecture slightly complex to understand
- 4 VGG16/VGG19: Extremely large size and slow inference

11. BEST PRACTICAL USAGE

- 1 For mobile systems: GoogleNet
- 2 For highest accuracy: ResNet
- 3 For transfer learning: VGG16 or VGG19
- 4 For beginners: AlexNet

12. FINAL RECOMMENDATION

- 1 Modern Computer Vision projects should prefer RESNET
- 2 Low resource environments should prefer GOOGLNET
- 3 Simple feature extraction tasks can use VGG16
- 4 AlexNet mainly useful for academic learning